

ZABU ASIA ENGINEERING ACADEMY

contact@zabu-asia.com | Singapore

PROFESSIONAL ENGINEERING INDUSTRY PREPARATION WORKSHOP

Foundation Program — Course Structure & Program Details

“Bridging Academic Knowledge and Real-World Engineering Practice”

Document Version 1.0 | 2025 | For Review & Discussion

40

Total Contact Hours

6

Training Modules

~15

Participants Max

25+

Years Trainer Experience

1. Program Overview

Program Name	Professional Engineering Industry Preparation Workshop
Organizer	ZABU Asia Engineering Academy ZABU Asia Engineering Solutions
Tagline	<i>“Bridging Academic Knowledge and Real-World Engineering Practice”</i>
Total Duration	40 Contact
Delivery Format	Face-to-Face Saturday Sessions Evening Classes (Optional)
Class Size	Small Group — Maximum 10-15 Participants Per Intake
Venue	Singapore (Location to be confirmed and communicated to registered participants)

2. Target Audience

This workshop is designed for the following participant profiles:

- Fresh Engineering Graduates — polytechnic diploma or university degree holders entering the professional engineering industries
 - Engineers Seeking Career Transition — professionals moving into mission critical engineering sectors
 - Engineers Preparing for Semiconductor, Pharmaceutical & Data Center Industries
 - Junior Engineers with 0–5 years of experience seeking structured industry knowledge enhancement
 - CAD / BIM / Drafting Personnel aspiring to transition into professional engineering roles
 - For Engineering Companies & Employers
- (This workshop is also suitable for companies looking to strengthen the technical competency of their junior engineers, design engineers, CAD/BIM personnel and project engineers.)

Program Intent

This program is structured as a Professional Development Workshop and Engineering Knowledge Enhancement Course for Phase 1. It is designed to bridge the gap between academic engineering education and the practical, industry-specific knowledge required to contribute effectively in Singapore's high-technology engineering environment.

3. Curriculum Summary

No.	Module Title	Contact Hours	Cumulative
1	Introduction to Mission Critical Industries	4 hrs.	4 hrs.
2	Engineering Fundamentals Refresher	8 hrs.	12 hrs.
3	Practical Design Engineering	10 hrs.	22 hrs.
4	Cleanroom Engineering	8 hrs.	30 hrs.

ZABU ASIA Engineering Solutions			CONFIDENTIAL
No.	Module Title	Contact Hours	Cumulative
5	Real Project Exposure	8 hrs.	38 hrs.
6	Career Development & Industry Readiness	4 hrs.	42 hrs.
TOTAL CONTACT HOURS		42 hrs.	

4. Detailed Course Modules

MODULE 1	Introduction to Mission Critical Industries	4 HOURS
	Semiconductor Industry Overview — wafer fabrication, fab types, Singapore landscape	
	Pharmaceutical Facilities Overview — sterile manufacturing, GMP environment, Singapore pharma sector	
	Data Center Industry Overview — hyperscale, colocation, Singapore as regional data hub	
	EPCM Project Lifecycle — Engineering, Procurement, Construction & Management phases	
	Engineering Consultancy vs Contractor Roles — career pathways, responsibilities, expectations	

MODULE 2	Engineering Fundamentals Refresher	8 HOURS
	HVAC Fundamentals — heating, ventilation, and air conditioning principles for engineers	
	Psychometrics Basics — dry bulb, wet bulb, dew point, enthalpy, humidity ratio	
	Airflow Principles — laminar vs turbulent flow, Bernoulli, pressure, velocity	

- Pump Fundamentals — pump curves, system resistance, affinity laws, NPSH
- Chilled Water Systems — primary/secondary circuits, delta T, coil selection
- Cleanroom Concepts — contamination control, filtration, pressurization overview

MODULE

3

Practical Design Engineering

10
HOURS

- Design Logic — how experienced engineers think through a design problem
- Engineering Calculations — hand calculations, spreadsheet tools, checking methodology
- Sizing Methodologies — duct sizing, pipe sizing, equipment selection logic
- Equipment Selection — reading vendor datasheets, selection criteria, technical submittals
- Reading Engineering Specifications — Division 23 HVAC specs, technical requirements
- Reading Engineering Drawings — P&IDs, HVAC layouts, sections, schedules
- Vendor Coordination — RFI process, technical clarification, submittal review

MODULE

4

Cleanroom Engineering

8
HOURS

- ISO Classification Standards — ISO 14644-1, particle count limits, at-rest vs in-operation
- Pressure Cascade Design — zone-by-zone pressurization, differential pressure values
- Air Change Rate Design — ACH calculation, diversity factors, ISO compliance

- Filter Fan Unit (FFU) Systems — HEPA/ULPA selection, coverage calculation, integrity testing
- Cleanroom HVAC Design — MAU, RAHU, recirculation systems, process exhaust

MODULE

5

Real Project Exposure

8

HOURS

- Semiconductor Facility Systems — fab floor layout, sub-fab, process tool connections, chemical exhaust
- Data Center Cooling Systems — CRAC/CRAH units, hot/cold aisle containment, PUE optimization
- Pharmaceutical HVAC Systems — Grade A–D requirements, EU GMP Annex 1 (2022), qualification
- Utility Systems Overview — compressed dry air, nitrogen, process cooling water, industrial gases
- Common Design Challenges — real project issues, lessons learned, how experienced engineers solve problems

MODULE

6

Career Development & Industry Readiness

4

HOURS

- Singapore Engineering Industry — key employers, EPCM firms, MNC client landscape
- Engineering Documentation — technical reports, calculation packages, design basis documents
- Interview Preparation — what firms look for, technical interview questions, how to present yourself
- Technical Presentation Skills — communicating engineering concepts clearly and professionally
- Engineering Career Pathway — progression from junior to senior engineer, professional development

5. Participant Outcomes & Deliverables

Upon successful completion of the workshop, each participant will receive:

- Workshop Participation Letter — issued by ZABU Asia Engineering Solutions
- Comprehensive Course Notes — full coverage of all 6 modules
- Engineering Quick Reference Guides — key formulas, standards, and design parameters
- Sample Calculations — worked examples for HVAC load calculations and sizing
- Engineering Templates — document templates for professional engineering use
- Industry Knowledge Resources — curated reference materials and standards overview
- Access to ZABU Asia free Engineering Knowledge Article Series (online)

6. Pricing Structure

Pilot Batch Pricing (Initial Intake):

Fee Category	Early Bird Rate	Standard Rate
General Participants	SGD 380	SGD 550
Group Registration (3 or more)	SGD 420	SGD 600
Myanmar Engineers (Pilot Batch)	SGD 380	SGD 450

** All fees are inclusive of course materials, printed notes, and Workshop Participation Letter.*

** Early Bird rate applicable to first 10 confirmed registrations per intake.*

7. Registration Process

To register or enquire about this program:

- Email: contact@zabu-asia.com

- Include in your email: Full name, contact number, current occupation / educational background, preferred fee category
- Seat confirmation and payment details will be provided within 2 working days
- Venue and exact schedule dates will be confirmed and communicated to all registered participants once the intake is finalized
- Maximum 10-15 participants per intake — early registration is strongly encouraged

ZABU ASIA ENGINEERING ACADEMY

"Together, we engineer solutions. Together, we build a better world."

contact@zabu-asia.com | Singapore